

Algebra II with Trigonometry

Chapter 5.2

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4401268_alg-2trig-h-chp-5-2-note-docx.pdf](#)

Based on your provided material, here are 10 *mental math* questions focused on core trig concepts. Each can be answered without paper, pencil, or calculator, testing reasoning and recall:

1.

What is the sine of 30° ?

2.

If the cosine of an angle in a right triangle is $1/2$, what is that angle in degrees?

3.

Which trigonometric ratio is always the reciprocal of sine?

4.

If one angle in a right triangle is 45° , what is the other non-right angle?

5.

What is $\tan(45^\circ)$?

6.

If $\sin(\theta) = 1/2$ and θ is an acute angle, what is $\cos(\theta)$?

7.

If the acute angle in a 30-60-90 triangle is 60° , what is the length of the side opposite this angle if the hypotenuse is 2?

8.

What is the value of $\sin(60^\circ) \times \cos(30^\circ)$?

9.

If the angle of elevation to the top of a building is 90° , where are you standing relative to the building?

10.

If $\tan(\theta) = 1$, what is θ in degrees between 0° and 90° ?

Let me know if you'd like the answers or explanations for these!